POOJA PATEL

+64210394058 | ppat965@aucklanduni.ac.nz | linkedin.com/in/965-ppatel | github.com/ppat965

PERSONAL STATEMENT

I am a recent Master of Engineering graduate majoring in Computer Systems Engineering, with a background in biomedical science and hardware and software design. My master's thesis focused on data engineering in a healthcare context, giving me an interdisciplinary approach to problem-solving. I thrive in roles with a people focus, whether as a census collector, in retail, or baking for a local charity. I am currently pursuing roles that aim to address real-world problems through technological innovation.

EDUCATION

Master of Engineering – Computer Systems | University of Auckland **JUL 2023 – JUL 2024**

Bachelor of Engineering (Hon) – Computer Systems | University of Auckland **FEB 2019 – NOV 2022**

Bachelor of Science – Biomedical Science | University of Auckland **FEB 2015 – JUL 2018**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, embedded C, HTML, CSS, JavaScript, R, SQL, MATLAB, and VHDL.
- **Frameworks/Libraries:** SpringBoot, Pandas, Scikit-learn, React, TensorFlow and PyTorch.
- **Tools:** Git, GitHub, Docker, REST APIs, and Visual Studio Code.
- **Cloud & DevOps:** Amazon Web Services.
- **Certifications:** AWS Certified Cloud Practitioner – Amazon Web Services (2026)

PROJECTS

Towards reliable asthma attack prediction using Machine Learning – Master's Project

- Engineered a custom **Python pipeline** using **Pandas** and **Scikit-learn** to automate **data preprocessing**, imbalanced dataset handling (e.g. **SMOTE/resampling**), and **hyperparameter tuning**.
- Used pipeline across multiple **Machine Learning** classifiers and **TensorFlow neural network** models.
- Conducted **feature extraction** and **correlation analysis** on clinical smart device data to identify biomarkers predating asthma attacks.

Pooja's Perspective – Web Portfolio

- Developed a static website deployed on **GitHub pages** using **Jekyll**, **HTML**, and **CSS**.
- Designed and implemented a custom theme with **adaptive** screen layout and am managing **version control** with **GitHub** (Link: ppat965.github.io).

Wireless Tracking System for Seabirds – Part 4 Project

- Built a **Python** simulation of **LoRa**-based bird tracking using **Two-Ray** and **Plane Earth** propagation models for **RSSI**-based **trilateration**.
- Modelled real-world conditions by incorporating **thermal noise** and **Monte Carlo** simulations.
- Proposed a 12-base station architecture across the 4,000km^2 Hauraki Gulf, achieving a mean **localisation** error of 101m and produced full **hardware** specifications for a physical tracker.

EXPERIENCE

Duty Manager – Foxton Food Market **DEC 2018 – AUG 2026**

- Manage a team of five across stock organisation, staff training and regulatory compliance, maintain a valid Duty Manager Certificate for the Horowhenua region.
- Identified an operational bottleneck in stock price updates and implemented a remote price change system, reducing the time spent on price updates by 50%.

Graduate Teaching Assistant – University of Auckland **JUL 2023 – NOV 2023**

- Supported lab sessions where students used Boolean logic to program a robotic car.
- Proactively identified struggling students and provided targeted assistance and mentorship, achieving 100% lab completion rate.

Census Collector – Stats NZ **FEB 2023 – MAY 2023**

- Engaged the diverse community across the region to promote census participation.
- Assisted throughout the completion process, contributing to a 91% response rate in the region.

Digital Solutions Intern – Department of Corrections NZ **NOV 2022 – FEB 2023**

- Presented a review of accessibility and inclusivity across technology at the company.
- Created a protocol for developers to assess accessibility in applications based on WCAG and NZ Web Standard guidelines using existing software.
- Audited internal application user guides, created a standardised template, and rewrote 25% of documentation to meet the new format, improving consistency across the product suite.

Teaching Assistant – University of Auckland **JUL 2022 – NOV 2022**

- Supervised over sixty students weekly on PSoC-based pathfinding robot project, debugging C/C++ algorithm code and fostering a positive learning environment.

ACHIEVEMENTS

- Best Part 4 Project – Telecommunication and Environmental Systems Category (2022)
- License Controller Qualification (2016)
- Runner-Up Dux of Manawatū College (2014)
- The successful applicant of The Rotary National Science and Technology Forum (2014)

VOLUNTEERING AND INTERESTS

- Volunteer – Pink Ribbon Appeal (2023, 2025), University Open Day (2023), Good B****** Baking (2025-)
- Subscriber – Service 95 Book club (2024-)
- Participant – Kiwi Jam (2022, Unity and C# game design)
- Class Representative for Physics 121 in Semester One, 2019

REFERENCES

REFEREES AVAILABLE ON REQUEST